Methods and Tips (24-72 hours)

While cold fermentation is a popular and convenient choice for making pizza dough, many bakers struggle with inconsistent results. Most issues arise from 'hybrid' fermentation methods – mixing room temperature and refrigeration – which introduce variables that are difficult to control. This guide introduces the Lehmann Method: a scientifically grounded, professional approach designed to produce perfectly fermented dough every time

Introduction to Cold Fermentation Management

When it comes to the cold fermentation of pizza dough, there are two primary fermentation 'methods' or dough management procedures:

1. Placing the dough in the fridge immediately after kneading and balling it (what I refer to as **the Lehmann method**).
2. **Hybrid fermentation**, which involves allowing the dough to ferment at room temperature before and/or after the time it spends in the fridge.

In the following sections, we will explore each method, discussing their benefits and/or drawbacks, as well as how to perform them properly.

It's important to note that both fermentation methods, and specifically the Lehmann method, work great for making **all types of pizzas**.

The Lehmann Method: Immediate Balling & Refrigeration

This is the fermentation method to be used with PizzaBlab's dough calculator and also the method I recommend for cold fermentation. Whether you are using the dough calculator to make dough for pizza or for other purposes, the principle remains the same: once you have finished kneading and portioning the dough, place it in the fridge as soon as possible.

This fermentation method was established by the late Tom Lehmann, also known as "The Dough Doctor". Tom was a food scientist, a renowned baking and dough consultant, and the former chairman of the American Institute of Baking.

Tom was one of the most influential mentors and experts in the field of pizza in the USA (and possibly worldwide). I personally learned a great deal from his extensive knowledge. His contribution to the world of pizza is immense and deserves

recognition above all else. This is my humble contribution to carrying on his legacy.

This fermentation method is my personal favorite and is also the preferred method used by the majority of pizzerias for cold fermentation. It is also the fermentation method to be used with PizzaBlab's dough calculator for cold fermentation (and we will soon understand why).

The primary purpose of this fermentation method is to provide predictable and consistent fermentation results, which are crucial in dough preparation, especially in a pizzeria or business setting. Hence, this method is preferred for cold fermentation in pizzerias. This is achieved by cooling the dough as quickly as possible after kneading and portioning/balling it.

The main advantage of this fermentation method is that it minimizes the factors that can influence the rate of fermentation (which we will discuss later). This creates an optimal and predictable fermentation process, allowing us to consistently achieve dough with excellent performance and a wide usability window.

Why the Lehmann Method Produces Superior Flavor, Texture and Consistency

In order to understand how cold fermentation using the Lehmann method helps achieve more consistent and predictable fermentation results, it is important to first understand what happens to the dough when it is refrigerated, especially when dealing with a large amount of dough (bulk fermentation).

For more information on this topic, refer to the following article: [Bulk Fermentation vs Ball Fermentation: Effects on Pizza Dough and Timing Guidelines]

In short:

1. The larger the mass of dough, the longer it takes for its interior to cool. When a large mass of dough is placed in the fridge, it can take up to 12 hours or more for the interior to reach the fridge's actual temperature. This means that **the dough will ferment at a much higher temperature than that inside the fridge**.
2. When the dough is left to ferment **at room temperature before being refrigerated**, the yeast produces CO_2, which forms an "insulating layer" within the dough. This layer slows down the cooling process of the dough's interior. The longer the dough ferments prior to refrigeration, the more CO_2 is produced, resulting in a larger insulating layer and a longer cooling time in the fridge.

3. The yeast generates heat as it works in the dough, further impeding the cooling of the dough's interior.

As can be seen, the combination of these factors means that dough that is cold fermented in bulk or allowed to ferment before being placed in the fridge will actually ferment at a higher temperature than the temperature inside the fridge. Consequently, the dough will ferment much faster, potentially resulting in unexpected fermentation outcomes, such as over-fermentation.

Using the Lehmann method allows us to effectively 'solve' the two issues mentioned above:

1. Instead of having one large mass of dough, we **divide the dough into smaller balls**. This allows for **quicker cooling compared to a larger mass of dough**. Think about what freezes faster in the fridge: a whole bottle of water or small ice cubes.
2. We "prevent" the yeast from starting to work and producing CO2 before placing the dough in the fridge. This prevents the formation of an insulating CO2 layer and allows for **faster cooling**.

By combining these two approaches, we achieve a dough that cools much faster once it is in the fridge. Consequently, the dough spends more time at the intended fermentation temperature and avoids being at a significantly higher temperature than planned, as is the case with hybrid fermentation (which we will discuss later).

If you prefer or need to ferment the dough in bulk (whether for space-saving reasons or otherwise), another option is to **flatten the dough in the container** to a thickness of no more than 2-3 cm (0.5-1"). This allows the dough to cool quickly, minimizing the impact of bulk fermentation on the fermentation temperature.

In essence, using the Lehmann method **allows us to ferment the dough in the fridge at a temperature that closely aligns with the intended fermentation temperature**. This yields fermentation results that are more **predictable** and **consistent**.

As we will explore later, hybrid fermentation often leads to the dough fermenting at a temperature significantly higher than intended, making it challenging to attain consistent and predictable fermentation results.

Another key advantage of the Lehmann method is its **simplicity**: you can refrigerate the dough immediately after kneading and use it almost straight from the fridge, eliminating the need for precise timing or additional fermentation steps:

1. Immediate refrigeration: The dough goes straight into the fridge after kneading, eliminating any waiting period. This prevents accidental over-fermentation at room temperature, ensuring consistent results.
2. Ready-to-use after refrigeration: The dough is fully fermented when removed from the fridge, requiring no additional fermentation time. Simply allow it to reach working temperature before use.

Yeast Calibration: Why Cold Fermentation [Using the Lehmann Method) Requires More Yeast

If you are accustomed to other cold fermentation methods, specifically hybrid fermentation (which we will discuss in the next section), you may find the yeast amount provided by the calculator to be very high compared to what you are used to.

The reason for this is simple: as we learned in the the lower the fermentation temperature, the more yeast is needed.

PizzaBlab's dough calculator was designed to be used with the Lehmann method for cold fermentation in order to achieve consistent and predictable results as much as possible. Therefore, **the yeast amount provided for cold fermentation is based on the assumption that the dough is divided into balls and immediately placed in the fridge after kneading**, so that it reaches the desired temperature as quickly as possible.

On the contrary, in hybrid fermentation, the internal temperature of the dough will be higher for a significant portion of the time in the fridge, so the yeast amount should be smaller accordingly.

Let's consider an example to illustrate this (the numbers used are arbitrary for demonstration purposes, but not necessarily far from reality):

We have two doughs: one fermented using the Lehmann method, and another fermented using hybrid fermentation (bulk fermented in the fridge). The total fermentation duration is assumed to be 24 hours in both cases, with a fridge temperature of 4C/40F degrees.

- The dough fermented using the Lehmann method is immediately placed in the fridge after kneading and dividing into balls. Out of the 24 hours in the fridge, each ball will have an internal temperature of 4C/40F for **20 hours**.
- The dough fermented using the hybrid method goes into the fridge as a single mass (bulk) after one hour of fermentation at room temperature. Out of the 24 hours in the fridge, the internal temperature of the dough will be 4C/40F for only **5 hours**. During the remaining 14 hours, it will actually ferment at a higher temperature (a temperature that cannot be accurately predicted).

The above example perfectly illustrates why it is necessary to use more yeast when using the Lehmann method. Fermentation at a **lower** temperature requires **more** yeast, while fermentation at a **higher** temperature (hybrid fermentation) requires **less** yeast.

This is why many calculators, which only calculate hybrid fermentation, provide a very small amount of yeast, and why PizzaBlab's dough calculator provides a relatively high amount of yeast.

It is important to note that using a higher amount of yeast is not inherently worse, and the "goal" of fermenting dough is not to use as little yeast as possible, but rather to adjust the amount of yeast according to the fermentation duration and temperature in order to achieve optimal fermentation. **The amount of yeast is a means to an end, not the ultimate goal**.

Theoretically, in both cases in the above example, it is possible to achieve the same level of flavor in the dough by adjusting the amount of yeast to different fermentation conditions (lower/higher fermentation temperature).

As long as the amount of yeast is adjusted to the duration and temperature of fermentation, a dough containing 0.5 grams of yeast is not 'better' or more flavorful than a dough containing 2 grams of yeast. This distinction is completely meaningless and should not be a consideration.

This is precisely why PizzaBlab's calculator provides a relatively high amount of yeast for cold fermentation. The amount is tailored to the Lehmann method and takes into account the actual fermentation at a low temperature, compared to other

calculators, which are specifically designed for hybrid fermentation.

Based on my experience and the experience of many others, using PizzaBlab's calculator's recommended amount of yeast in combination with the Lehmann method yields optimal fermentation results for cold fermentation.

Implementation: A Practical Guide to the Lehmann Method

1. Knead and Control FDT:
Knead/mix the dough as usual, aiming for a Final Dough Temperature (FDT) of **15–20°C / 60–70°F**. This lower starting point allows the dough to reach the fridge's target temperature faster, ensuring a more predictable fermentation rate.

2. Immediate Balling:
Once kneading is complete, immediately **divide the dough into balls**. (If you are only making a single pizza, you can skip this step).

Note: Depending on your kneading method, you may need to let the dough rest at room temperature for 10–20 minutes before shaping (but no more than that). Intensive kneading can result in a very elastic dough at the end of mixing, which can resist balling. A short rest allows the gluten to relax, making the dough softer, more extensible, and easier to handle.

3. The 20-Minute Rule:
Place the dough in the fridge **within 20 minutes** of finishing kneading. Let it ferment for your intended duration. If your workflow requires folds, you can briefly remove the dough from the fridge, perform the folds, and return it to the fridge.

4. Balling / Reballing / Pan Stretching:
If your process involves reballing, if you used the flattened bulk method and need to ball the dough, or if you need to stretch the dough into a pan for pan pizzas, remove the cold dough from the fridge and shape your balls (or stretch into the pan) as usual, in accordance with your fermentation schedule.

Note: You may need to let the dough rest at room temperature for 15-30 minutes before shaping/stretching; otherwise, the cold dough may be too elastic and resist shaping.

5. Monitoring fermentation:
The dough is ready to be baked when it has roughly doubled in volume while in the fridge.

Note: At standard refrigeration temperatures (1–5°C / 34–40°F), yeast and enzyme activity slows down significantly but does not stop. Consequently, the dough continues to both ferment and mature, just at a decelerated rate.

If you are using the Lehmann Method and observe no visible sings of fermentation (such as a volume increase or bubbles along the bottom and sides of the container) it indicates a flaw in the process (e.g., incorrect yeast quantity, an FDT that was too low, or a fridge that is too cold). When following this method correctly, the dough **should** double in volume during its time in the fridge.

6. Using the Dough (Warm-up Period):

At the end of the fermentation period, remove the dough from the fridge. Let the balls rest **at** room temperature until their internal temperature reaches at least **10°C / 50°F**. This usually takes 30 minutes to 3 hours.

Note: You are not waiting for the dough to reach room temperature; you are simply allowing it to warm up enough to facilitate easier stretching. While the dough is technically ready to bake straight from the fridge, cold dough is highly elastic and resistant. Resting it until it hits **10°C, 15°C, or 20°C** (depending on your preference) ensures the gluten is relaxed and extensible enough to stretch into a pizza base. **Crucially, this rest is for physical handling, not to allow for additional fermentation.**

Style-Specific Dough Temperature Considerations

The **internal** temperature of your dough at the moment of baking can impact the final crust, depending on the style of pizza you are making:

- **Neapolitan Pizza:** For high-heat environments (e.g., wood-fired ovens at 430°C+ / 800°F+), it is recommended to let the dough reach a higher internal temperature, around 20°C / 68°F. Baking cold dough at these extreme temperatures can cause aggressive "leoparding".
- **Hand-Stretched Pizzas:** For hand-stretched pizzas baked in standard electric or gas ovens (such as NY style), the dough can be worked at an internal temperature as low as 10°C / 50°F, provided you find it extensible enough to shape comfortably.
- **Pan Pizzas (Detroit style, Sicilian, etc.):** Because these doughs are stretched into the pan before the final proof, they can often be moved directly from the fridge to the oven. Since no manual shaping is required at the time of baking,

there is no need to wait for the dough to warm up – it can be baked cold straight from the fridge.

Understanding Hybrid Fermentation Methods

The Mechanics of Combining Room and Cold Fermentation

Hybrid fermentation is a method that combines cold fermentation with room temperature fermentation. The principle behind this method is simple: the dough is fermented at room temperature for a certain period of time **before** (and after) being refrigerated for the remaining fermentation time.

The key aspects of this fermentation method include:

1. After kneading, the dough is allowed to ferment at room temperature for an hour or more to "activate" the yeast.
2. The dough is then transferred to the fridge for the desired fermentation duration (24-48-72 hours).
3. A few hours before baking, the dough is taken out of the fridge to complete the fermentation process at room temperature. If the dough was fermented as a bulk, it is divided into balls after removing it from the fridge and allowed to ferment further before baking.

In the hybrid fermentation method, a **very small amount of yeast** is usually used. There are two reasons for this:

1. The dough should undergo additional fermentation **at room temperature**. The fridge is used to slow down the fermentation process – the dough is not very active during its time in the fridge, primarily because of the relatively low amount of yeast.
2. This fermentation method causes the dough to ferment in the fridge at a temperature higher than the actual fridge temperature, which means less yeast is needed.

These two reasons result in the need for a relatively small amount of yeast (usually between 0.1-0.3% instant dry yeast), which is suitable for the fermentation conditions of this method.

Due to the small amount of yeast, dough undergoing hybrid fermentation **does not show significant signs of fermentation while in the fridge**. This is why it is important to allow the dough to "complete" its fermentation after removing it from

the fridge. **The amount of yeast in the dough is too low to cause significant rising and maturation activity in the fridge**, but enough to cause these activities when the dough is fermented at room temperature.

At low temperatures, the CO2 produced by the yeast (which is what causes the dough to gain volume) becomes much more soluble in water, transitioning from a gas to a liquid state.

This is one of the reasons why dough that undergoes cold fermentation, with a small amount of yeast, and specifically hybrid fermentation, often shows no significant physical change in volume during its time in the fridge. The concentration of CO2 in the dough is low (due to the small amount of yeast), and most of the CO2 is in a liquid state rather than a gas, preventing the dough from physically rising.

When the dough is removed from the fridge, the liquid CO2 becomes less soluble and reverts back to a gaseous state. This contrasts with the Lehmann method, in which a larger amount of yeast produces more CO2, leading to visible signs of fermentation in the dough even during the refrigeration period.

In other words, in hybrid fermentation, the fermentation time **outside** the fridge can sometimes be more important than the time spent in the fridge in terms of maturation and flavor development.

That's why in hybrid fermentation it's crucial to allow the dough to "finish" fermenting outside the fridge before baking, unlike the Lehmann method where the dough is ready to be baked after the fermentation time in the fridge.

In fact, it can even be said that the flavors in dough undergoing hybrid fermentation primarily come from the time **outside the fridge**, as the contribution of the time in the fridge to flavor development is smaller than commonly believed, although it still exists.

It's important to note that not every dough undergoing hybrid fermentation will necessarily follow the same pattern. The results depend on specific temperatures and other variables, which is why this fermentation method can yield inconsistent and less predictable results compared to the Lehmann method, as we'll see in the next section.

Like any fermentation method, the hybrid fermentation method works well when done correctly. The challenge, however, is that consistently executing it correctly is not a simple task. Personally, I believe in simplifying the process as much as

possible, because **why work harder when you can work smarter?**

Why Hybrid Methods Lead to Inconsistent Fermentation

So, what is the main drawback of the hybrid fermentation method? In short, **dough that undergoes hybrid fermentation will ferment much less consistently and predictably, and is much more susceptible to external influences** such as the room temperature before and after being in the fridge, the time spent outside the fridge, and the final dough temperature.

The longer the fermentation time in the fridge, the more significant these effects will be.

Why does this happen? I have already discussed this in previous sections, but to summarize:

1. The yeast activity in the dough produces carbon dioxide (CO2) or "gas bubbles" that make the dough less dense and more airy. These gas bubbles act as a physical "insulation layer" inside the dough, slowing down the transfer of heat or cold between the environment and the dough. **Less compressed dough (more airy) means it is better insulated from the environment**.
2. **The metabolic activity of the yeast itself generates heat** (about 0.5°C per hour). Depending on the degree of insulation from the environment, this heat may remain inside the dough and further affect its internal temperature.
3. **The larger the mass of dough, the longer it will take for the external temperature to have an effect on it, and the effect will occur more slowly**. For example, compare the melting time of a small ice cube to that of a shoebox-sized block of ice.

The above process is a kind of "cycle" – as the dough ferments (becoming more airy and creating a larger insulation layer), it is less affected by the ambient temperature. At the same time, the yeast's activity generates more heat in the dough (which is retained due to the insulation layer). This heat, in turn, increases yeast activity and accelerates fermentation (more heat equals higher yeast activity). As a result, the insulation layer continues to grow, thus perpetuating the cycle.

What does this mean in practice? The longer we let the dough ferment at room temperature before refrigerating it, and the higher the temperature and/or the larger the dough (in terms of weight and volume), and/or the higher the final dough temperature, the "thicker" the "insulation layer" will become. As a result, the dough will be better able to maintain its internal temperature.

In other words, when we place such dough in the fridge, it already has an insulating layer that prevents its core from reaching the temperature of the fridge. As a result, **the dough ferments at a higher temperature than intended**, which leads to **unpredictable** and **inconsistent** fermentation results.

In practice, this means that **controlling the fermentation of dough fermented using the hybrid method is much more challenging**. It depends on several factors that are not easy (or even possible) to account for, even when using various calculators.

This is why PizzaBlab's calculator does not offer an option for hybrid fermentation. Even if a certain amount of yeast is obtained for hybrid fermentation, it cannot be relied upon (the same applies to other calculators that calculate yeast amounts).

I had the opportunity to see (and taste) many pizzas that people made using the hybrid fermentation method, and they were highly satisfied with the results. However, upon observing and tasting pizzas made by the same person at different times using a supposedly identical process (hybrid fermentation), there is a noticeable lack of consistency in the final product. At times, the dough ferments perfectly, while at other times it is over-fermented or under-fermented. This inconsistency is often a result of hybrid fermentation.

On top of that, the fact that the individual dough balls need to undergo additional fermentation after being taken out of the fridge can lead to potential problems. The time it takes for the balls to be ready for baking varies greatly depending on factors such as temperatures and the amount of yeast used.

This requires us to allow the dough balls to undergo another fermentation "step", which also involves (physical) rising and, most importantly, ensuring that the dough "relaxes" and becomes extensible enough to be easily stretched into a pizza base without being too elastic and resistant (a relatively common case with hybrid fermentation).

This is in contrast to the Lehmann method, where the dough balls are ready for baking in terms of fermentation and maturation once they are taken out of the fridge. Using the Lehmann method, we simply need to wait for the dough balls to reach working temperature, without the need for another fermentation stage.

When using hybrid fermentation, the speed at which the above processes occur (the dough balls rise enough and become extensible enough) greatly depends on room temperature, the amount of yeast used, and the extent of fermentation before and

during the time spent in the fridge. In other words, this method can be highly inconsistent and unpredictable, making it challenging to determine when the dough will be ready for baking after being taken out of the fridge.

In conclusion, while the hybrid fermentation method is relatively common for fermenting pizza dough, **it is much less consistent than the Lehmann method**, especially for those who are just starting out or have less experience in controlling and assessing the fermentation process.

Let it be clear – this method can work excellently with the right process and the proper amount of yeast. However, the problem lies in the fact that making these adjustments is not necessarily easy, and achieving consistent results becomes much more difficult as a result.

If you've ever heard statements like "In the winter, you need to allow the dough to rest for X hours outside the fridge, and in the summer for Y hours", it illustrates why the hybrid fermentation method is inconsistent compared to the Lehmann method. It complicates matters and is highly influenced by external factors that are not necessarily within our control, which inevitably affects the final product.

Best Practices for Successful Cold Fermentation

Fridge Microclimates: Optimizing Temperature Stability

Usually, the temperature inside the fridge varies at different points, and the location of the dough in the fridge can greatly impact the rate of fermentation.

In general:

- The temperature at the **back** of the fridge is lower.
- The shelves at the same level as the cooling unit (usually the upper shelves) have a lower temperature due to the cold air coming out.

It's important to note that there can be significant temperature differences, ranging from 1°C/34°F to 7°C/44°F, that can noticeably affect yeast activity and the rate of fermentation. To accurately measure the temperature in a specific location inside the fridge, use a temperature sensor, and make sure you place the dough in the same spot to achieve consistent fermentation results.

Condensation in the Fermentation Container

On the right: condensation on the lid of a container holding cold-fermented dough

Condensation is the process through which water vapor in the air transforms into water droplets when it encounters a colder surface. When we place dough in a closed container in the fridge (which is always at a lower temperature than the container), the hot air inside the container comes into contact with the sides of the container (which cools down faster than the dough itself).

As a result, the air cools and loses its ability to hold moisture. Consequently, the excess moisture in the air condenses, **forming water droplets on the lid and sides of the container** (as shown in the picture above).

In summary, the temperature difference between the dough and the fridge causes condensation and the formation of water droplets in the container (particularly on the lid), which then drip back onto the dough.

Condensation always occurs when room temperature dough is placed in a closed container inside the fridge.

When the dough is **bulk fermented** in the fridge, condensation is generally not a significant issue, as the dough will later be divided and shaped into balls, allowing any surface moisture to be “reincorporated” into the dough.

However, when the dough ferments in the fridge as **individual balls**, water droplets dripping onto the dough can lead to several negative effects:

- The drops of water can make the dough sticky. The water may remain on the surface or be absorbed unevenly over time, creating **wet spots**.
- During baking, the water in these wet spots evaporates, potentially causing **large bubbles** to form, either on the pizza base or around the crust.

- When stretching the dough into a pizza base, dusting flour can stick to these wet areas, leading to **flour buildup**. During baking, this excess flour may burn and create a bitter taste.
- The dough may become **stickier and harder to handle**, increasing the risk of it sticking to the work surface or pizza peel.

Managing Moisture and Condensation in the Fridge

To prevent condensation in the container, it is necessary to **allow the dough to cool down in the fridge until it reaches an internal temperature of at least 10°C/50°F before closing or sealing the container**.

To accomplish this, simply place the container with the dough in the fridge **partially covered**, and let it sit like this for approximately one to two hours. Once the dough has sufficiently cooled (you can check its internal temperature with a thermometer), you can then close or seal the container and continue with the fermentation process as usual. To prevent the upper part of the dough from drying out while it is in the fridge uncovered, lightly spread oil on top of the dough balls (any type of oil will work).

Alternately, you can also place a **damp towel** over the container, which will prevent hot air from entering, and help absorb any water droplets caused by condensation.

Alternatively, you can place the dough in one of the closed drawers in the refrigerator, such as the vegetable or "fresh" drawer.

Cooling Techniques: How to Rapidly Reduce Dough Temperature

As we have discussed in the previous sections, it is important to cool the dough in the fridge as quickly as possible for effective cold fermentation. This is key to achieving a controlled, consistent, and predictable fermentation process, with a wider usability window.

In addition to the methods we have already covered (specifically using the Lehmann method), here are some additional tips for speeding up the cooling process, especially if you choose to bulk ferment dough in the fridge rather than as individual balls. These tips can help you achieve a more uniform fermentation, with a greater usability window:

1. Divide a large mass of dough into several smaller portions and use different containers for each portion. Similar to dividing the dough into balls, dividing it

into separate portions will result in faster cooling compared to having one large dough mass.

2. Flatten the dough inside the containers as much as possible. A flat(ter) dough will cool faster than a spherical one, thanks to its larger surface area-to-volume ratio.
3. Aim for a lower final dough temperature of 15-20°C/60-70°F. This lower temperature will help facilitate faster cooling of the dough.
4. Use a fermentation container with high heat conductivity, such as aluminum. This will enable the cold from the refrigerator to transfer to the dough more quickly.
5. Place the dough in the coldest spot of the fridge, ideally close to the vents from where the cold air comes out. Usually, this spot is at the back of the top shelf. To be certain, you may need to explore different areas of your refrigerator.

Key Takeaways

- **The Lehmann Method:**
  This method involves dividing the dough into balls and placing them in the fridge **immediately** after kneading. This professional standard minimizes external variables and ensures the dough ferments at a stable, predictable temperature from start to finish.
- **The Hybrid Challenge:**
  Hybrid fermentation (letting dough sit at room temperature before or after refrigeration) introduces unpredictable temperature fluctuations. This makes the fermentation rate nearly impossible to track accurately.
- **Superior Flavor Development:**
  Unlike hybrid methods that use tiny amounts of yeast (causing the dough to "hibernate" in the fridge), the Lehmann Method uses enough yeast to allow the dough to **actively ferment** at cold temperatures. This continuous maturation throughout the cold phase results in a significantly superior flavor development compared to hybrid fermentation methods.
- **PizzaBlab's Dough Calculator:**
  PizzaBlab's Dough & Yeast Calculator is specifically calibrated for the Lehmann Method, providing the exact yeast quantities needed to achieve a full rise within your cold-fermentation window.
- **Ball vs. Bulk:**
  Dividing dough into balls before refrigerating allows for faster, more even cooling. Large bulk masses "trap" internal heat, causing the center to ferment faster than the exterior, leading to inconsistent dough.

- **Yeast Logic:**
  The Lehmann Method requires **more yeast** than hybrid methods. Because the dough cools rapidly, the yeast must be present in high enough concentrations to remain active and functional at lower temperatures.
- **The Working Window:**
  Dough fermented using the Lehmann method is ready when it has roughly doubled in volume. You don't need to "finish" it at room temperature – simply let it rest until it reaches at least 10°C (50°F) for optimal extensibility before stretching.
- **FDT Control:**
  A lower Final Dough Temperature (15–20°C / 60–70°F) is recommended for cold fermentation. This gives the dough a "head start" on cooling, allowing it to reach fridge temperature faster and making the fermentation rate much more predictable.